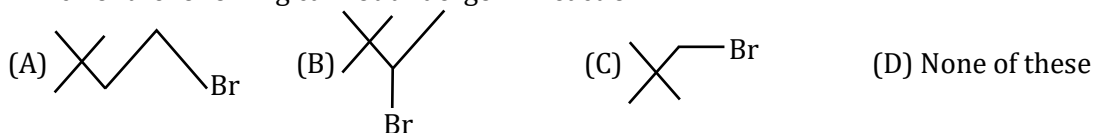


EXERCISE (O-1)

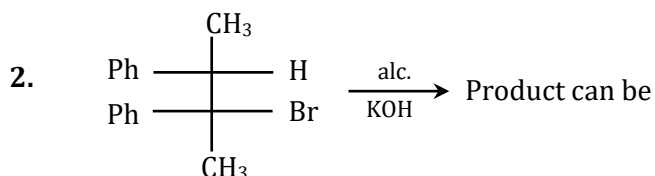
SINGLE CORRECT TYPE QUESTIONS

Elimination reaction (E2) [Base like alc.KOH, NaNH₂, RLi, t-BuOK]

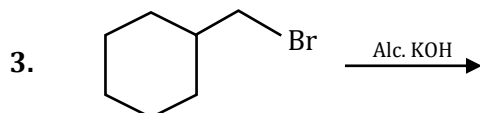
1. Which of the following cannot undergo E2 reaction?



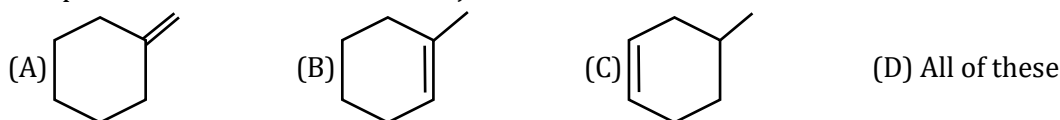
CRM001



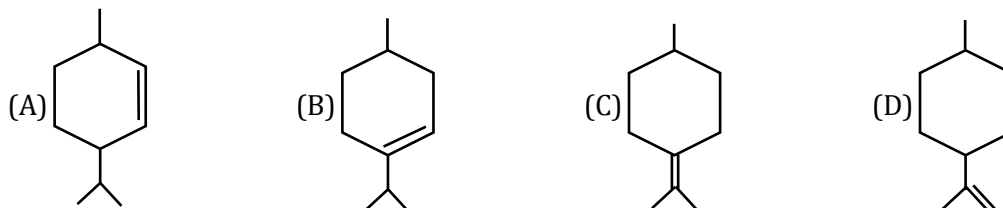
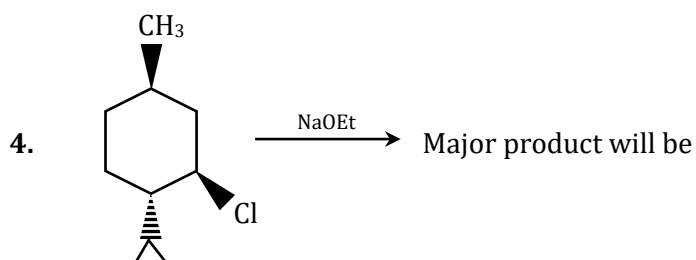
CRM002



The product obtained when it is subjected to E2 reaction will be

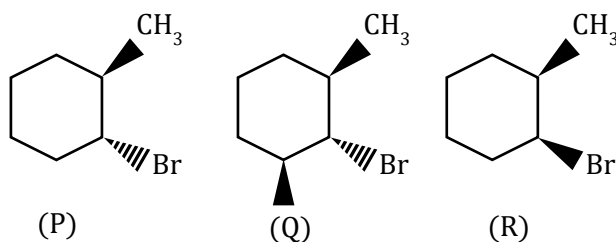


CRM003



CRM004

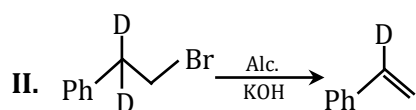
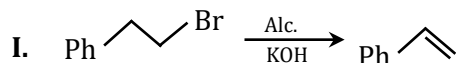
5. Arrange the following in order of decreasing rate of E2 reaction



- (A) $P > Q > R$ (B) $R > P > Q$ (C) $R > Q > P$ (D) $Q > P > R$

CRM005

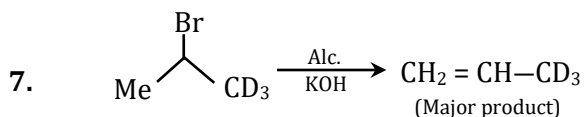
6. In the following reactions:



The rate of reaction of (I) and (II) are same. Both reactions proceed by which mechanism.

- (A) E1 (B) E2 (C) E1 cB (D) Anti-elimination

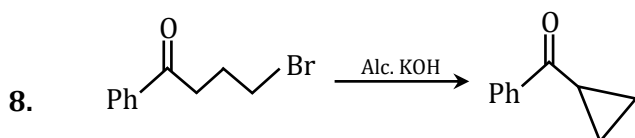
CRM006



By which mechanism does the above reaction proceed?

- (A) E1 (B) E2 (C) E1 cB (D) γ -Elimination

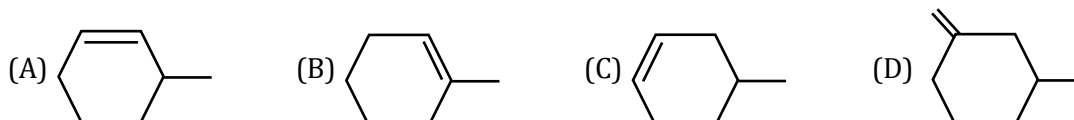
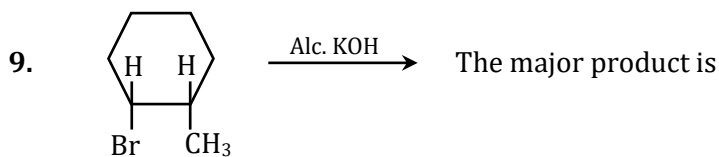
CRM007



By which mechanism does the above reaction proceed?

- (A) α -Elimination (B) β -Elimination (C) γ -Elimination (D) δ -Elimination

CRM008



CRM009

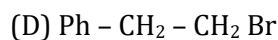
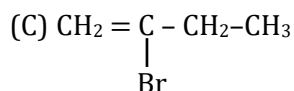
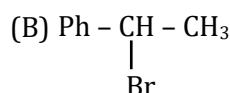
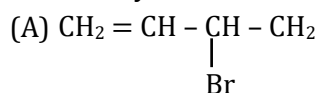
Elimination reaction (E1)

10. Which product is formed in E1 reaction of 1-Bromo-2, 3-dimethylcyclohexane?

- (A) 3, 4-Dimethylcyclohexane (B) 2, 3-Dimethylcyclohexane
(C) 4, 5-Dimethylcyclohexene (D) 1, 2-Dimethylcyclohexene

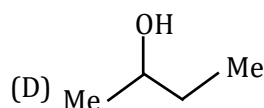
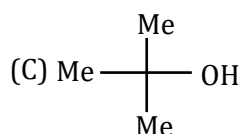
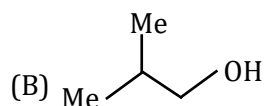
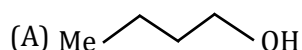
CRM010

11. Which alkyl halide would you expect to be least reactive in E1 reaction ?



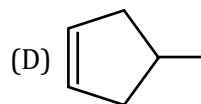
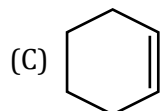
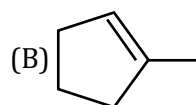
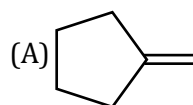
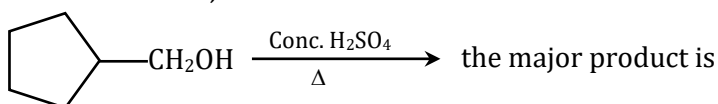
CRM011

12. The compound which would give the most stable carbocation on dehydration is:



CRM012

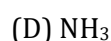
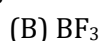
13. For the reaction,



CRM013

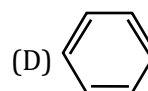
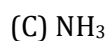
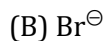
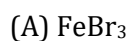
Nucleophilic Substitution reaction ($\text{S}_{\text{N}}2$) [Nucleophile such as KSH , RSK , AgNO_2 , KNO_2 , AgCN , KCN , NaNO_2 , Carbanion (R^-), NaX , aq.KOH , ROK , RNH_2]

14. Which of the following is not an electrophile?



CRM014

15. Which of the following is not a nucleophile?



CRM015

16. How are basicity and leaving group ability related?

(A) They are not related to each other

(B) Good leaving groups are strong bases

(C) Good leaving groups are weak bases

(D) Leaving group $\propto$ Basic strength

CRM016

17. Ethers can act as :

(A) Bronsted acids

(B) Bronsted bases

(C) Lewis acids

(D) Lewis bases

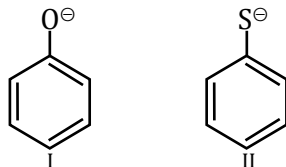
CRM017

24. Which is the MOST nucleophilic in the following series?

- (A) $F^{\ominus}$ (B) CH_3CH_2OH (C) H_2O (D) $CH_3CH_2O^{\ominus}$

CRM024

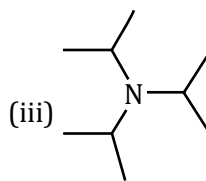
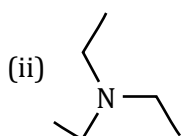
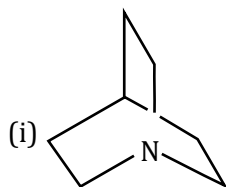
25. Consider the following two anionic molecules. Which of the following statements is TRUE ?



- (A) I is more basic and more nucleophilic in PPS than II.
 (B) I is less basic and less nucleophilic in PPS than II.
 (C) I is more basic but less nucleophilic in PPS than II.
 (D) I is less basic but more nucleophilic in PPS than II.

CRM025

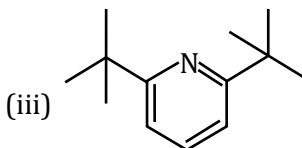
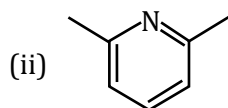
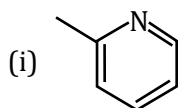
26. Identify correct nucleophilicity order



- (A) (i) > (ii) > (iii) (B) (ii) > (iii) > (i) (C) (i) > (iii) > (ii) (D) (iii) > (i) > (ii)

CRM026

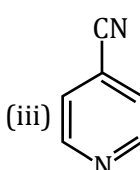
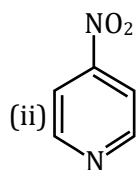
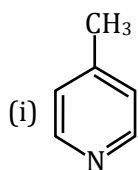
27. Identify correct nucleophilicity order



- (A) (i) > (ii) > (iii) (B) (ii) > (iii) > (i) (C) (i) > (iii) > (ii) (D) (iii) > (i) > (ii)

CRM027

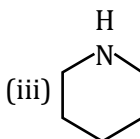
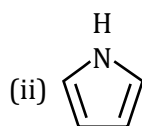
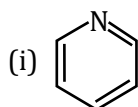
28. Identify correct nucleophilicity order



- (A) (i) > (ii) > (iii) (B) (ii) > (iii) > (i) (C) (i) > (iii) > (ii) (D) (iii) > (i) > (ii)

CRM028

29. Identify correct nucleophilicity order



- (A) (i) > (ii) > (iii) (B) (ii) > (iii) > (i) (C) (i) > (iii) > (ii) (D) (iii) > (i) > (ii)

CRM029

30. Identify correct nucleophilicity order

- (i) H_2O (ii) H_2S (iii) H_2Se
 (A) (i) > (ii) > (iii) (B) (ii) > (iii) > (i) (C) (i) > (iii) > (ii) (D) (i) > (iii) > (ii)

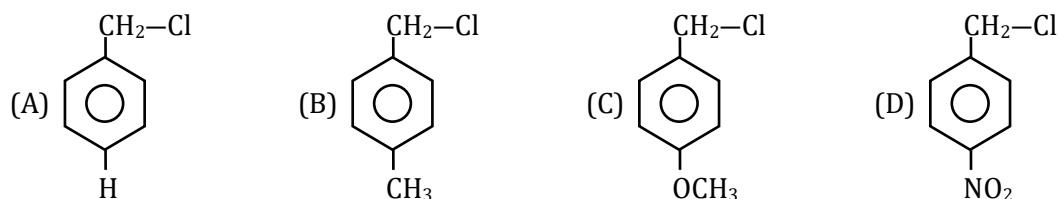
CRM030

31. Identify correct nucleophilicity order

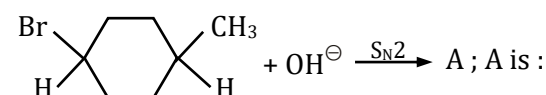
- (i) PH_3 (ii) NH_3 (iii) AsH_3
 (A) (i) > (ii) > (iii) (B) (ii) > (iii) > (i) (C) (i) > (iii) > (ii) (D) (ii) > (iii) > (i)

CRM031

32. Which of the following is most reactive toward $\text{S}_{\text{N}}2$.



CRM032

33.  A is :



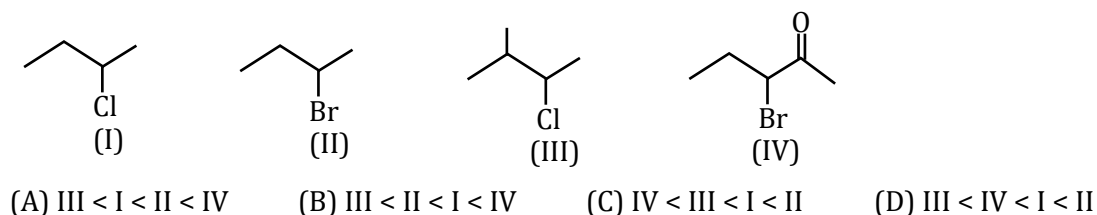
CRM033

34. The reactivity of 2-bromo-2-methylbutane (I), 1-bromopentane (II) and 2-bromopentane (III) towards $\text{S}_{\text{N}}2$ displacement is such that :

- (A) I > II > III (B) I > III > II (C) II > III > I (D) II > I > III

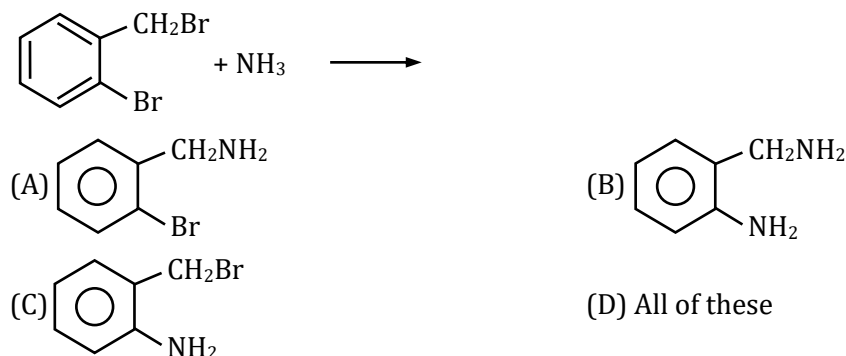
CRM034

35. Arrange these compounds in order of increasing $\text{S}_{\text{N}}2$ reaction rate :



CRM035

36. The major product in the given reaction:



CRM036

37. When ethyl bromide is treated with moist Ag_2O , the main product is:

- (A) Ethyl ether (B) Ethanol
(C) Ethoxy ethane (D) All of these

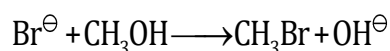
CRM037

38. Which of the following is most reactive for $\text{S}_\text{N}2$ reactions.

- (A) $\text{C}_6\text{H}_5(\text{CH}_3)(\text{C}_6\text{H}_5)\text{CBr}$ (B) $\text{C}_6\text{H}_5\text{CH}(\text{CH}_3)\text{Br}$
(C) $(\text{C}_6\text{H}_5)_3\text{CBr}$ (D) $\text{C}_6\text{H}_5\text{CH}_2\text{Br}$

CRM038

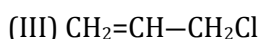
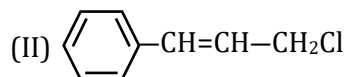
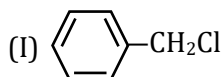
39. Select suitable reason for non-occurrence of the following reaction ?



- (A) attacking nucleophile is stronger one
(B) leaving group is strong base than nucleophile
(C) Alcohols are not good substrate for S_N reaction.
(D) Hydroxide ions are weak bases

CRM039

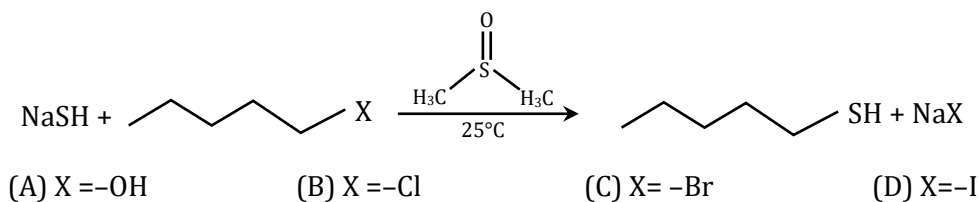
40. Arrange the following in the order of their reactivity of $\text{S}_\text{N}2$ reaction



- (A) (IV) > (III) > (I) > (II)
(B) (II) > (I) > (III) > (IV)
(C) (I) > (II) > (IV) > (III)
(D) (III) > (II) > (I) > (IV)

CRM040

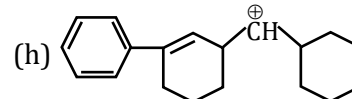
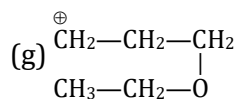
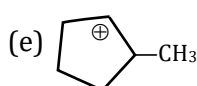
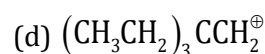
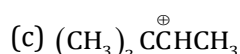
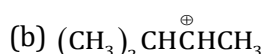
41. For which leaving group X would you predict the $\text{S}_\text{N}2$ reaction shown below to take place most rapidly?



CRM041

Nucleophilic Substitution reaction ($\text{S}_\text{N}1$)

42. How many following carbocation undergo re-arrangement –



(A) 5

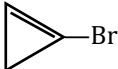
(B) 8

(C) 6

(D) 7

CRM042

43.  Br, which is not the correct statement:

- (I)
 (A) I is more soluble in water than bromocyclopropane
 (B) I gives pale yellow ppt. on addition with aq. AgNO_3
 (C) I is having lower dipole moment than bromocyclopropane
 (D) I is more ionic than 

(I)

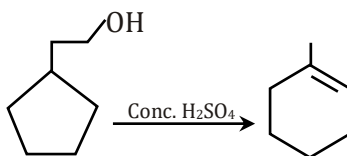
CRM043

44. A solution of (–)-1-chloro-1-phenylethane in toluene racemises in the presence of a small amount of SbCl_5 , due to the formation of :-

- (A) carbanion (B) Carbene
 (C) Carbocation (D) free radical

CRM044

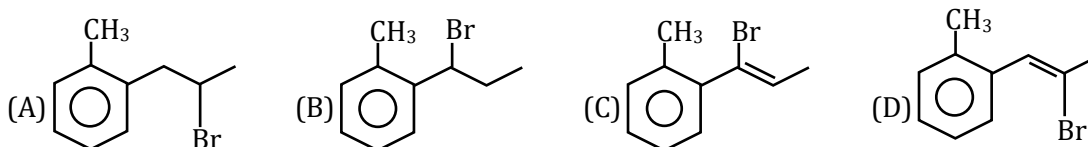
45. How many 1,2-Shifts of carbocation intermediate are involved during the course of following reaction :



- (A) 1 (B) 2 (C) 3 (D) 4

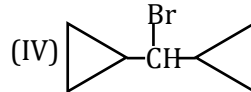
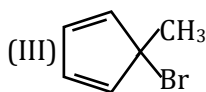
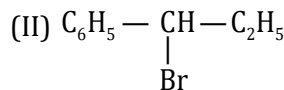
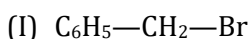
CRM045

46. Which compound undergoes hydrolysis by the $\text{S}_{\text{N}}1$ mechanism at the fastest rate?



CRM046

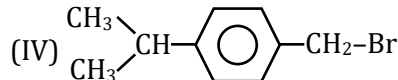
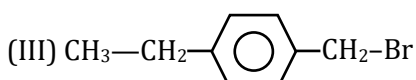
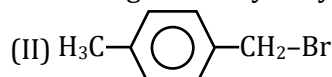
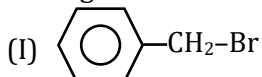
47. Arrange the following compounds in decreasing order of their reactivity for hydrolysis reaction



- (A) $\text{II} > \text{III} > \text{IV} > \text{I}$ (B) $\text{IV} > \text{II} > \text{I} > \text{III}$ (C) $\text{III} > \text{IV} > \text{II} > \text{I}$ (D) $\text{I} > \text{II} > \text{III} > \text{IV}$

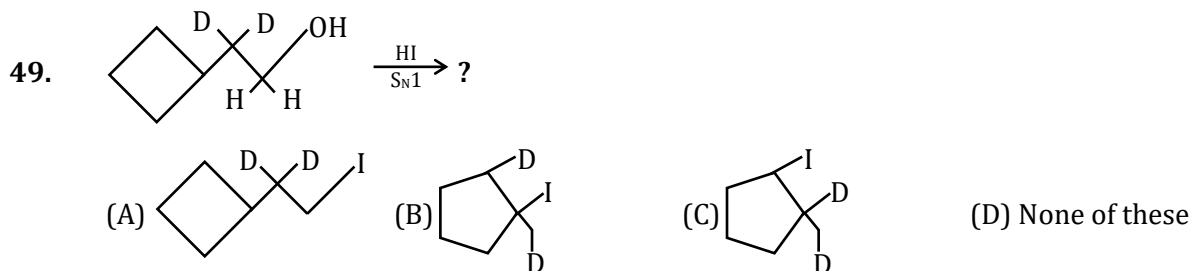
CRM047

48. Arrange the following compounds in order of decreasing rate of hydrolysis for $\text{S}_{\text{N}}1$ reaction:



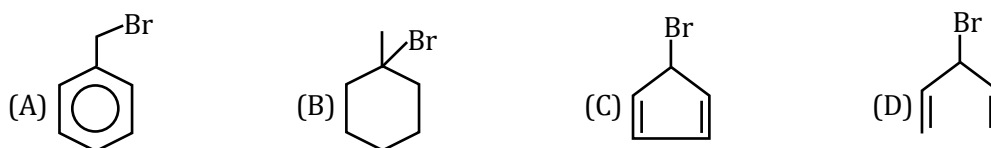
- (A) $\text{II} > \text{III} > \text{IV} > \text{I}$ (B) $\text{IV} > \text{III} > \text{II} > \text{I}$ (C) $\text{III} > \text{IV} > \text{II} > \text{I}$ (D) $\text{I} > \text{II} > \text{III} > \text{IV}$

CRM048



CRM049

50. Which of the following can not give $\text{S}_{\text{N}}1$ reaction easily?



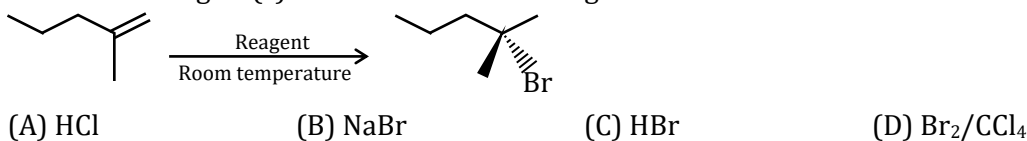
CRM050

51. Which one of the following compounds will be most reactive for $\text{S}_{\text{N}}1$ reactions?



CRM051

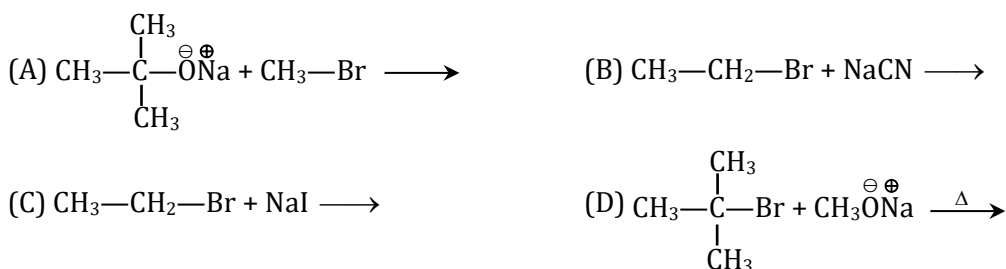
52. What is the reagent(s) needed for the following reaction?



CRM052

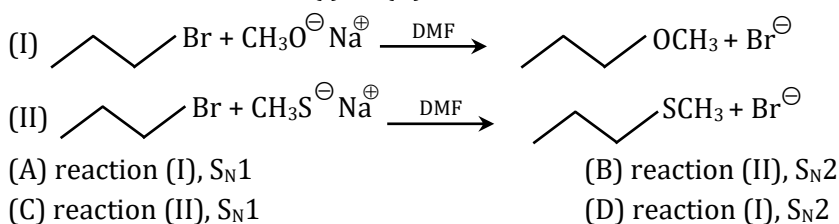
Nucleophilic Substitution reaction vs Elimination reaction (S_{N} vs E)

53. Which reaction gives elimination as a major product



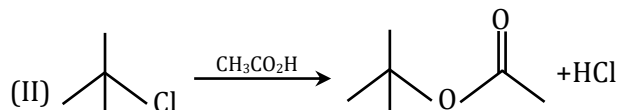
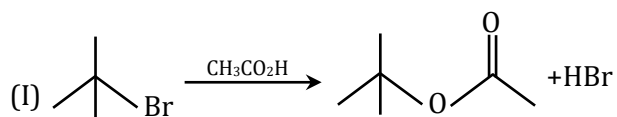
CRM053

54. Which reaction is faster (I) or (II) and what is its mechanism?



CRM054

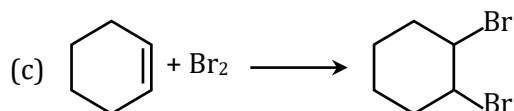
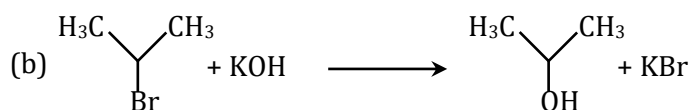
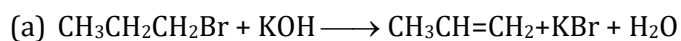
55. Which reaction is faster (I) or (II) and what is its mechanism?



(A) reaction (II), S_N1 (B) reaction (II), S_N2 (C) reaction (I), S_N1 (D) reaction (I), S_N2

CRM055

56. For the following reactions :-



Which of the following statement is correct?

- (A) (a) and (b) are elimination reaction and (c) is addition reaction
(B) (a) is elimination, (b) is substitution and (c) is addition reaction
(C) (a) is elimination, (b) and (c) are substitution reactions
(D) (a) is substitution, (b) and (c) are addition reaction

CRM056

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

Elimination reaction (E2) [Base like alc.KOH, NaNH₂, RLi, t-BuOK]

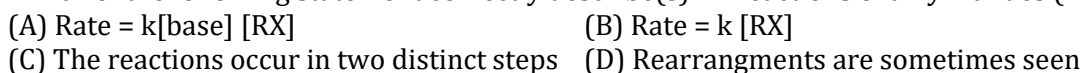
1. Which of the following is expected to give more than 2 products in E2 reaction?



CRM057

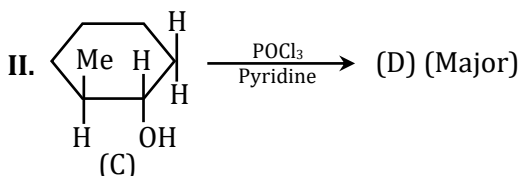
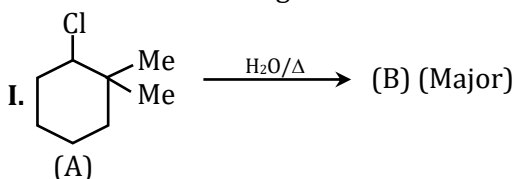
Elimination reaction (E1)

2. Which of the following statement correctly describe(s) E1 reactions of alkyl halides (RX)?



CRM058

3. Consider the following reaction:

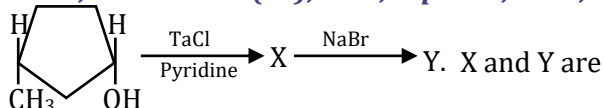


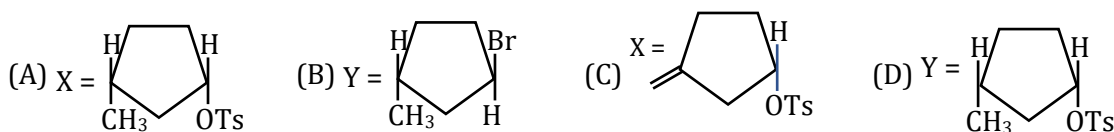
Which of the following statements are correct?



CRM059

Nucleophilic Substitution reaction (S_N2) [Nucleophile such as KSH, RSK, AgNO₂, KNO₂, AgCN, KCN, NaNO₂, Carbanion (R⁻), NaX, aq.KOH, ROK, RNH₂]

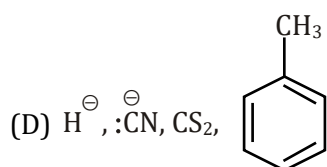
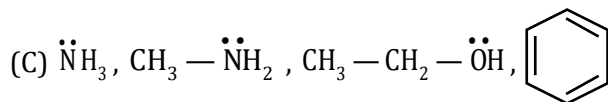
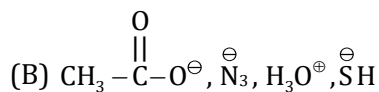
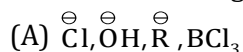
4.  X and Y are



CRM060

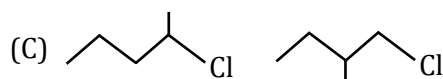
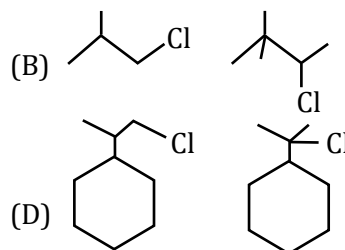
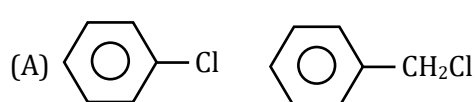
Reaction Mechanism

5. Which of following is/are not a set of nucleophiles?



CRM061

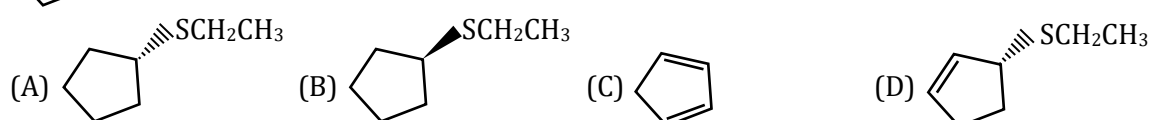
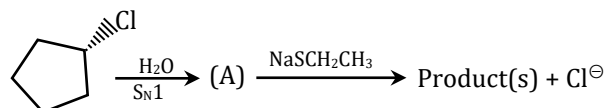
6. In the given pair in which pair the first compound is more reactive than second towards $\text{S}_\text{N}2$ reaction.



CRM062

Nucleophilic Substitution reaction ($\text{S}_\text{N}1$)

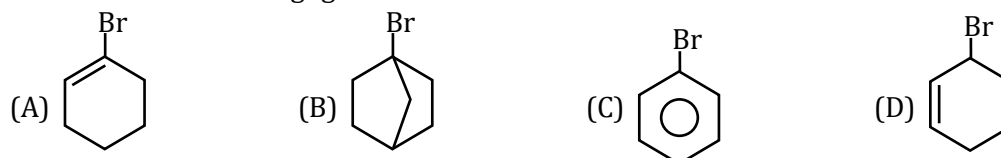
7. What could be the product(s) for the following $\text{S}_\text{N}1$ reaction?



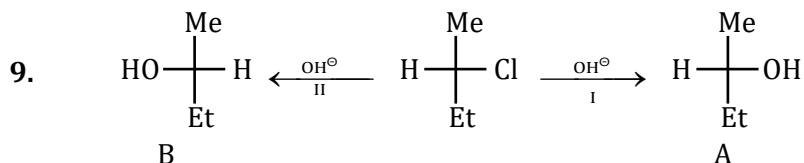
CRM063

Nucleophilic Substitution reaction vs Elimination reaction (S_N vs E)

8. $\text{S}_\text{N}2$ reaction will be negligible in



CRM064



Mechanism involved is/are:

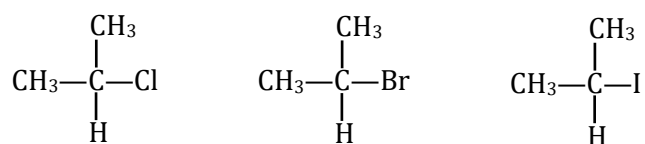
- (A) I can't be $\text{S}_\text{N}1$ (B) II can't be $\text{S}_\text{N}2$ (C) I is $\text{S}_\text{N}1$ (D) II is $\text{S}_\text{N}2$

CRM065

10. Incorrect statement among the following is/are :
- (A) R-OH with NaI in the presence of phosphoric acid gives R—I, but not in the absence of phosphoric acid
- (B) 2-methyl propane on chlorination ($\text{Cl}_2, h\nu$) gives 1-chloro-2-methyl propane while bromination ($\text{Br}_2, h\nu$) gives 2-bromo-2-methyl propane
- (C) Usually higher temperature prefers substitution over elimination
- (D) Triphenyl chloromethane cannot be hydrolysed

CRM066

11. From left to right, correct statements are :



- (A) Rate of $\text{S}_{\text{N}}1$ mechanism increases in polar protic solvent
- (B) Rate of $\text{S}_{\text{N}}2$ mechanism increases in DMSO
- (C) Rate of E_2 mechanism increases
- (D) Rate of E_1 mechanism increases

CRM067

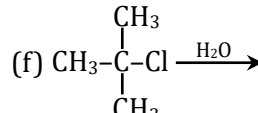
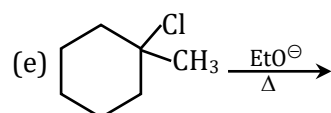
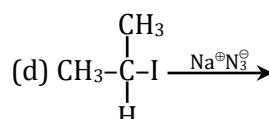
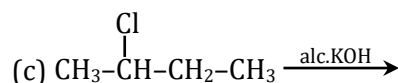
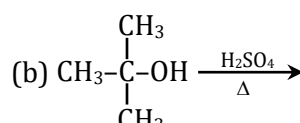
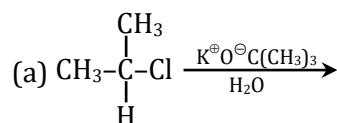
EXERCISE (S)

Part-1 : JEE Main (Numerical Value type)

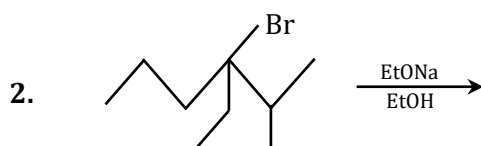
- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

Elimination reaction (E2) [Base like alc.KOH, NaNH₂, RLi, t-BuOK]

1. How many of the following reagent combinations will result in elimination reactions leading to the major product?



CRM068



How many total alkenes can be made?

CRM069

3. 2-Bromopentane + $\text{EtO}^\ominus \xrightarrow{\Delta}$ The total products are

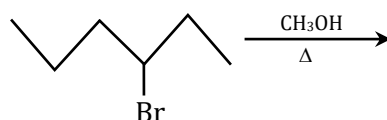
CRM070

Elimination reaction (E1)

4. How many major products can be formed by E1 reaction of 1-chloro-2,2-diphenyl propane ?

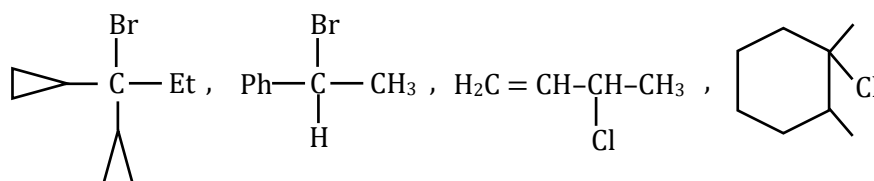
CRM071

5. Find number of possible E1 products from following reaction :



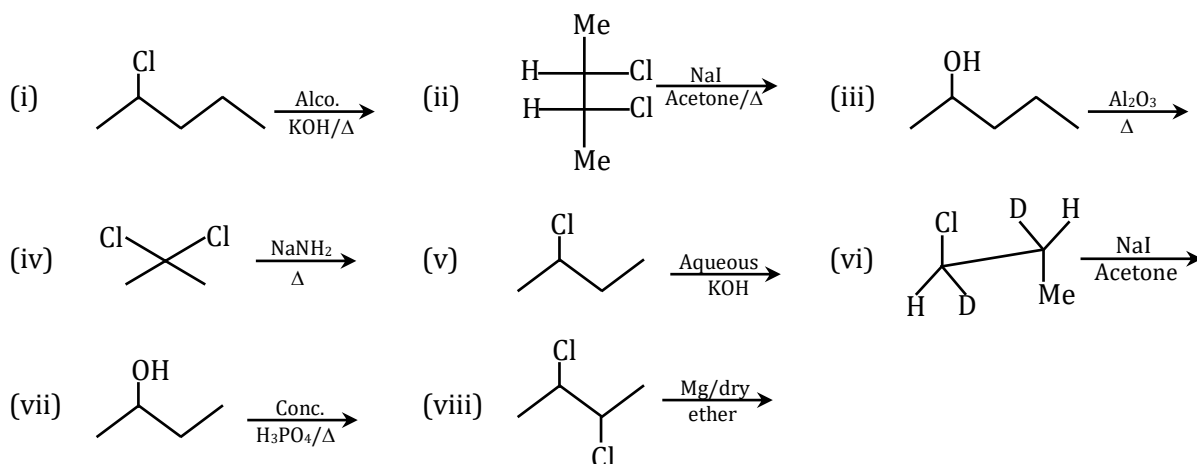
CRM072

6. How many of the following compounds will give E1 reaction?



CRM073

7. Identify total number of ' β '-elimination reactions



CRM074

8. Calculate total number of alkene products when 2-chloro-2-cyclobutyl hexane react with alcoholic KOH and heat.

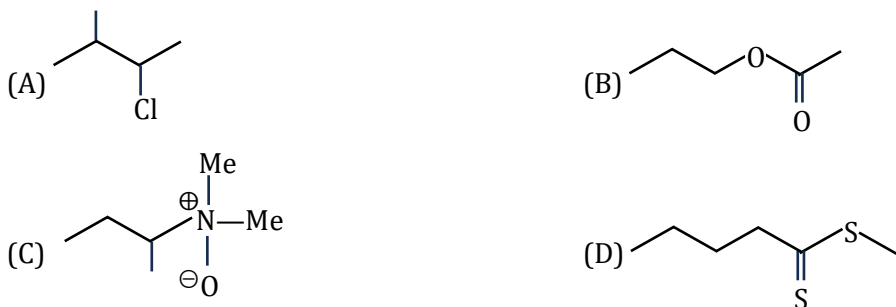
CRM075

9. Calculate total number of α -H present in major product alkene formed when 2,3-dimethyl butanol react with concentrated $\text{H}_2\text{SO}_4/\Delta$.

CRM076

Elimination reaction (Ei) [Ester, Xanthates, Amino oxides, Sulfoxide, Selenoxide]

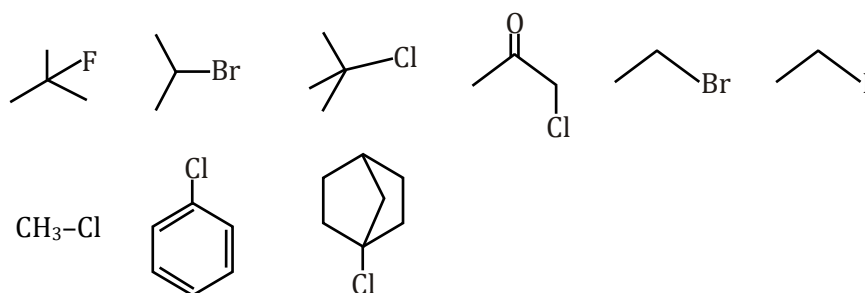
10. Number of following compounds will give syn elimination reaction.



CRM077

Nucleophilic Substitution reaction ($\text{S}_{\text{N}}2$) [Nucleophile such as KSH, RSK, AgNO_2 , KNO_2 , AgCN, KCN, NaNO_2 , Carbanion (R^-), NaX, aq.KOH, ROK, RNH_2]

11. Number of compounds which are more reactive than $\text{H}_3\text{C}-\underset{\text{CH}_3}{\text{CH}}-\text{Cl}$ towards $\text{S}_{\text{N}}2$ reaction:



CRM078

12. If 1,1-dichloro propane is reacted with aqueous NaNO_2 solution, only one of the chlorine undergo nucleophilic substitution ($\text{S}_{\text{N}}2$) reaction. How many mono-substitution products would be formed? (including stereomers)

CRM079

13. How many of the following compounds encourages " $\text{S}_{\text{N}}2$ " mechanism?

(A)		(B)		(C)	
(D)		(E)		(F)	

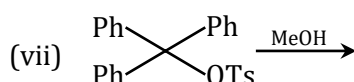
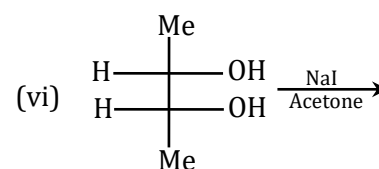
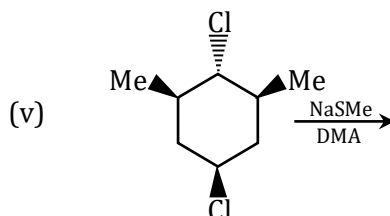
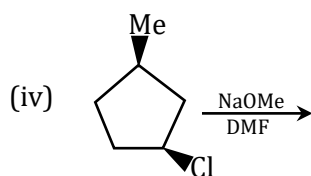
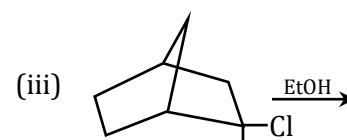
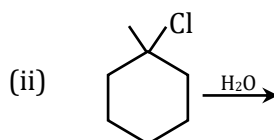
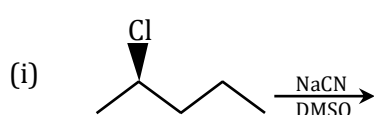
CRM080

Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

Nucleophilic Substitution reaction ($\text{S}_{\text{N}}1$)

14. Identify total number of $\text{S}_{\text{N}}1$ reactions

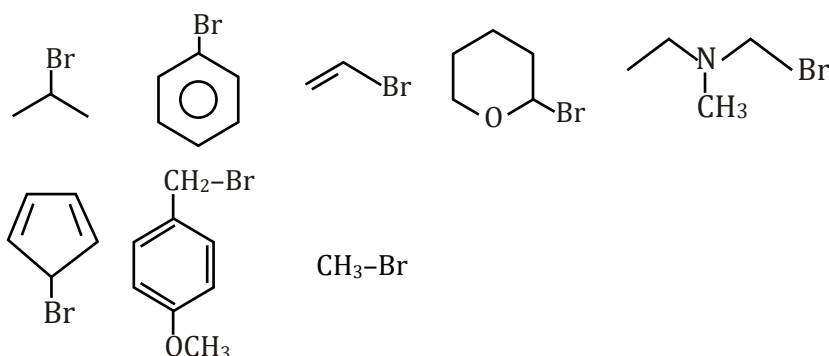


CRM081

15. Calculate percentage of $\text{S}_{\text{N}}1$ product if (R)-2-chloro butane on reaction with $\text{NaOH}/\text{H}_2\text{O}$ and acetone gives 98 per cent inverted product.

CRM082

16. Consider following compounds:



No. of compounds which are more reactive than CC(C)Cl for S_N1 reaction are :

CRM083

17. Number of the correct statement among the following

- (i) S_N1 reactions involve two step
- (ii) S_N2 reactions involve single step
- (iii) S_N1 reactions involves carbocation intermediate
- (iv) S_N2 reaction involves carbonium ion intermediate

CRM084

Nucleophilic Substitution reaction vs Elimination reaction (S_N vs E)

18. When (S)-2-chloropentane reacts with NaOH in 75 percent EtOH and 25 percent acetone follows rate law :

$$\text{Rate} = 2.35 \times 10^{-5} \left[\text{(S)-2-chloropentane} \right] \cdot [\text{OH}] + 5.30 \times 10^{-6} \left[\text{2-chloropentane} \right]$$

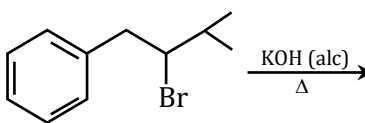
Then, calculate percentage of S_N1 product when concentration of $[\text{OH}] = 1.5$ molar.

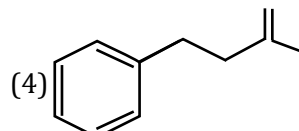
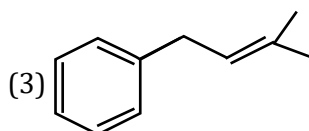
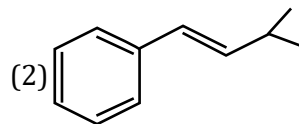
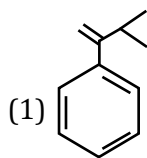
If percentage of S_N1 product = X, then calculate value of Y if.

$Y = X - 5$. (nearest integer)

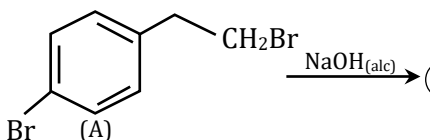
CRM085

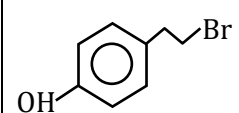
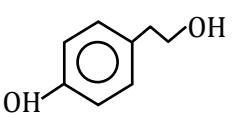
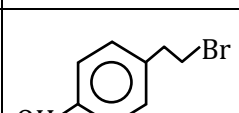
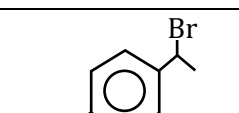
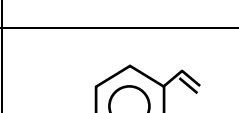
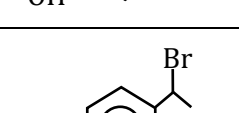
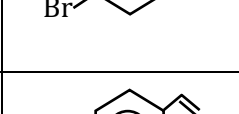
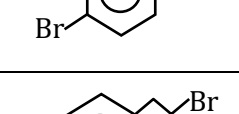
EXERCISE - JEE (Main) PYQ

1.  Major product "P" Product P is [JEE (Main) 2024]



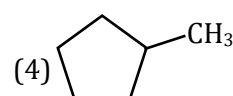
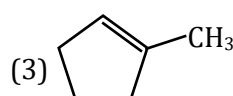
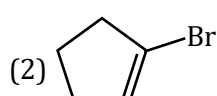
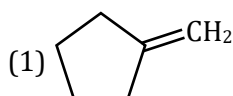
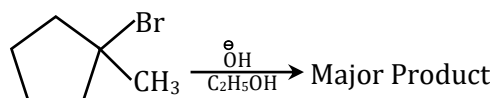
CRM086

2.  Identify B and C and how are A and C related ? [JEE (Main) 2024]

(1)			functional group isomers
(2)			Derivative
(3)			position isomers
(4)			chain isomers

CRM087

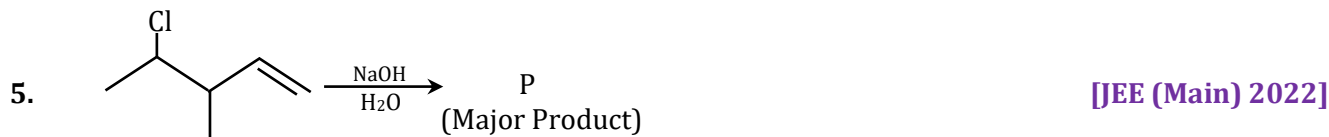
3. Identify the major product in the following reaction. [JEE (Main) 2024]



CRM088

4. The one giving maximum number of isomeric alkenes on dehydrohalogenation reaction is (excluding rearrangement) [JEE (Main) 2023]
 (1) 1-Bromo-2-methylbutane (2) 2-Bromopropane
 (3) 2-Bromopentane (4) 2-Bromo-3,3-dimethylpentane

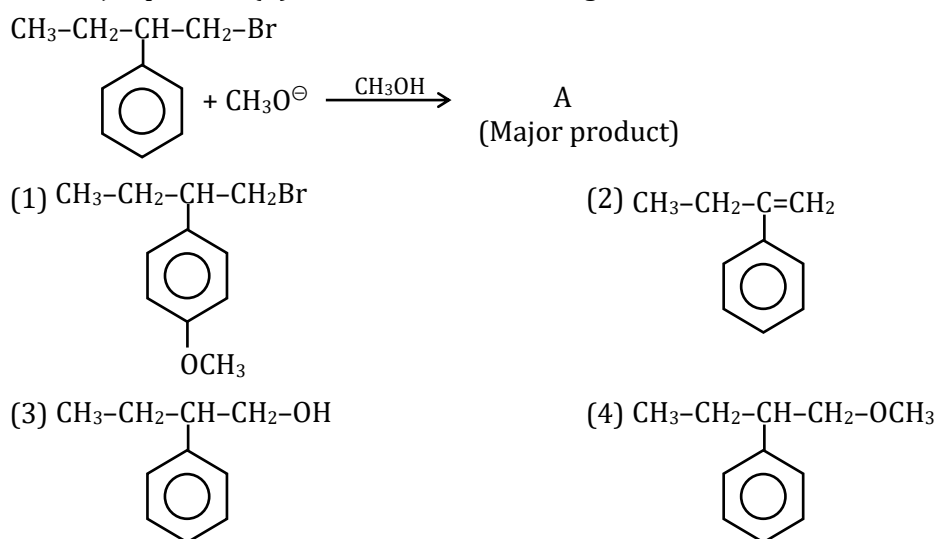
CRM089



Consider the above reaction. The number of π electrons present in the product 'P' is ____.

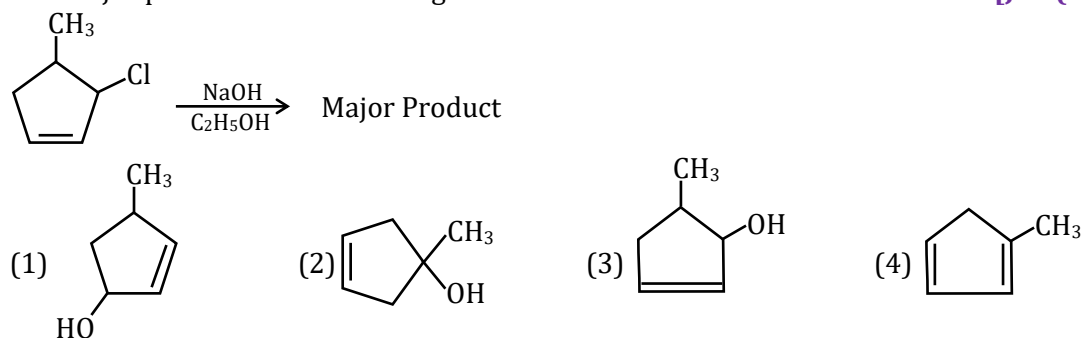
CRM090

6. The major product (A) formed in the reaction given below is : [JEE (Main) 2021]



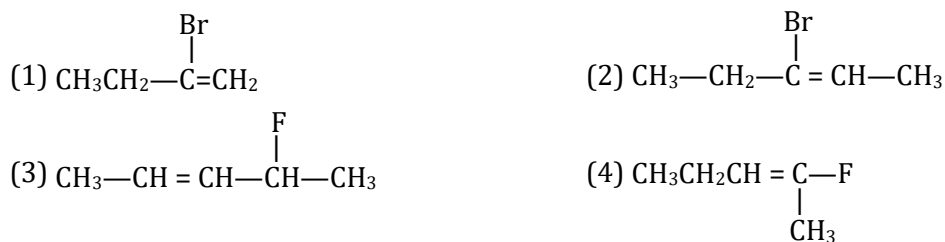
CRM091

7. The major product of the following reaction is : [JEE (Main) 2021]



CRM092

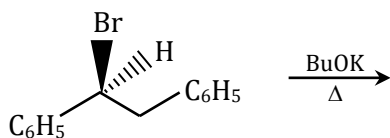
8. The major product obtained from E_2 -elimination of 3-bromo-2-fluoropentane is: [JEE (Main) 2020]



CRM093

9. The major product obtained in the following reaction is :-

[JEE (Main) 2017]

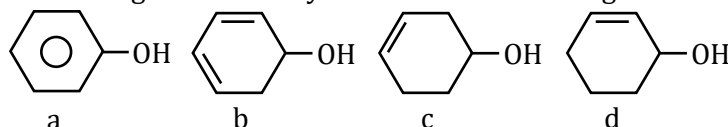


- (1) $(\pm)\text{C}_6\text{H}_5\text{CH}(\text{O}^t\text{Bu})\text{CH}_2\text{CH}_2\text{C}_6\text{H}_5$ (2) $\text{C}_6\text{H}_5\text{CH}=\text{CHC}_6\text{H}_5$
 (3) $(+)\text{C}_6\text{H}_5\text{CH}(\text{O}^t\text{Bu})\text{CH}_2\text{C}_6\text{H}_5$ (4) $(-)\text{C}_6\text{H}_5\text{CH}(\text{O}^t\text{Bu})\text{CH}_2\text{C}_6\text{H}_5$

CRM094

10. Decreasing order of dehydration of the following alcohols is

[JEE (Main) 2023]

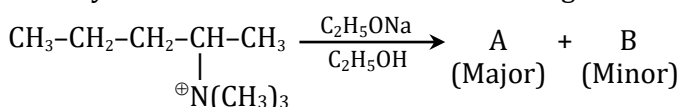


- (1) $a > d > b > c$ (2) $b > d > c > a$
 (3) $b > a > d > c$ (4) $d > b > c > a$

CRM095

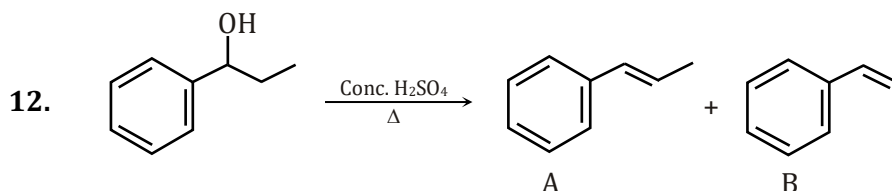
11. Identify the correct statement for the below given transformation.

[JEE (Main) 2022]



- (1) A - $\text{CH}_3\text{CH}_2\text{CH}=\text{CH--CH}_3$, B - $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}=\text{CH}_2$, Saytzeff products
 (2) A - $\text{CH}_3\text{CH}_2\text{CH}=\text{CH--CH}_3$, B - $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}=\text{CH}_2$, Hofmann products
 (3) A - $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}=\text{CH}_2$, B - $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_3$, Hofmann products
 (4) A - $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}=\text{CH}_2$, B - $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_3$, Saytzeff products

CRM096



Consider the above reaction, and choose the correct statement :

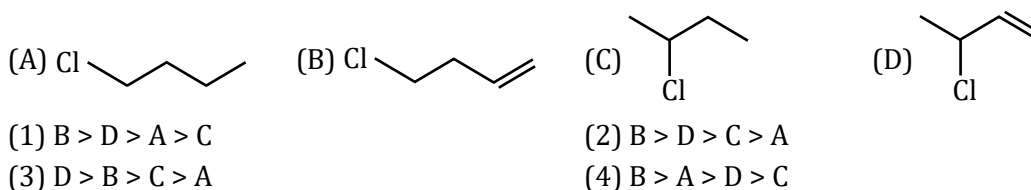
[JEE (Main) 2021]

- (1) The reaction is not possible in acidic medium
 (2) Both compounds A and B are formed equally
 (3) Compound A will be the major product
 (4) Compound B will be the major product

CRM097

13. The decreasing order of reactivity towards dehydrohalogenation (E_1) reaction of the following compounds is :

[JEE (Main) 2020]

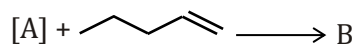
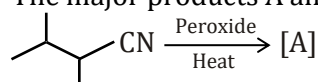


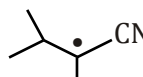
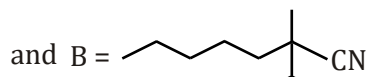
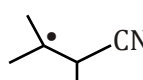
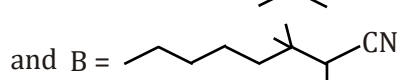
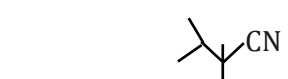
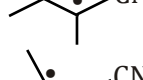
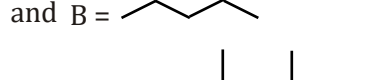
- (1) $B > D > A > C$ (2) $B > D > C > A$
 (3) $D > B > C > A$ (4) $B > A > D > C$

CRM098

14. The major products A and B in the following reactions are

[JEE (Main) 2020]

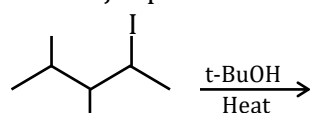


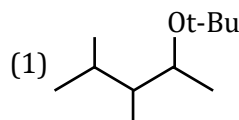
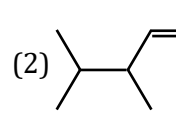
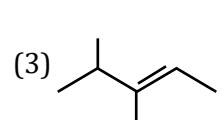
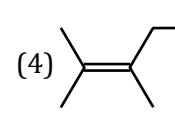
- (1) A =  and B = 
 (2) A =  and B = 
 (3) A =  and B = 
 (4) A =  and B = 

CRM099

15. The major product in the following reaction is :

[JEE (Main) 2020]

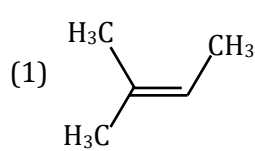
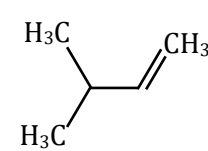
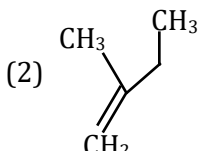
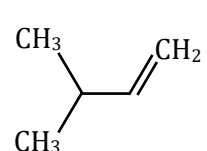
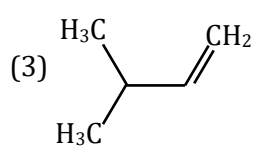
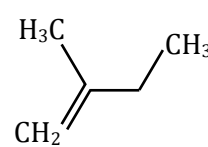
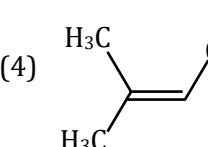
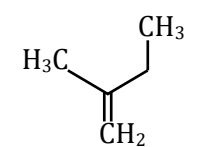


- (1)  (2)  (3)  (4) 

CRM100

16. When neopentyl alcohol is heated with an acid, it slowly converted into an 85 : 15 mixture of alkenes A and B, respectively. What are these alkenes ?

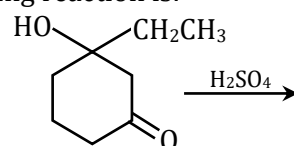
[JEE (Main) 2020]

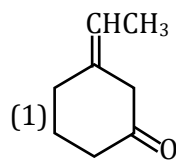
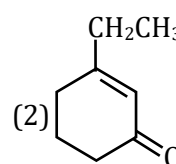
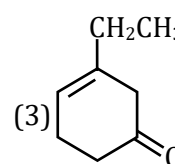
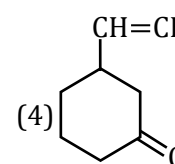
- (1)  and  (2)  and 
 (3)  and  (4)  and 

CRM101

17. The major product of the following reaction is:

[JEE (Main) 2020]

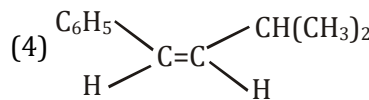
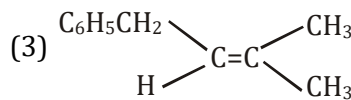
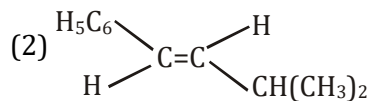
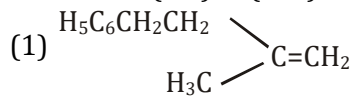
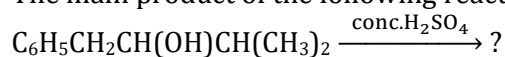


- (1)  (2)  (3)  (4) 

CRM102

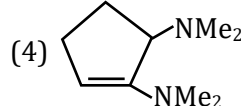
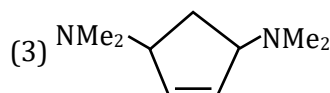
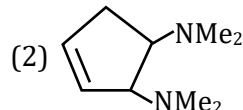
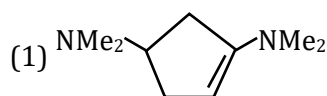
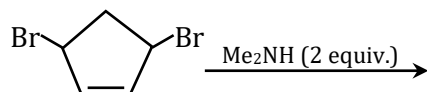
18. The main product of the following reaction is

[AIEEE-2010]



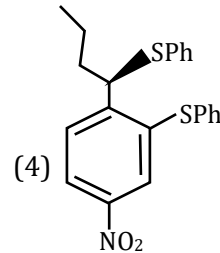
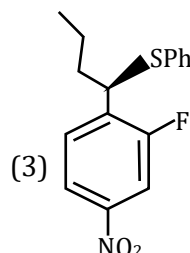
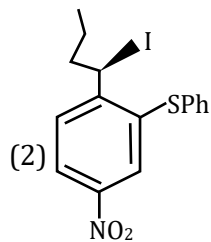
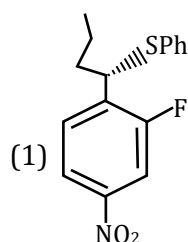
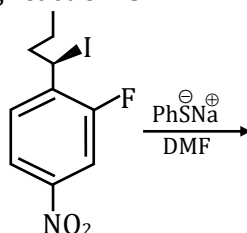
19. Find out the major product formed from the following reaction. [Me: -CH₃]

[JEE (Main) 2024]



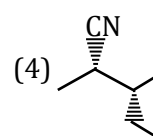
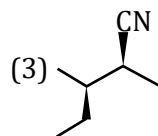
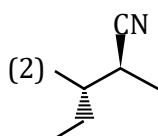
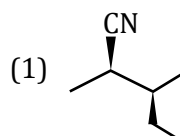
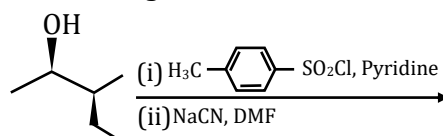
20. The major product of the following reaction is:

CRM104
[JEE (Main) 2022]



21. Most stable product of the following reaction is:

[JEE (Main) 2022]

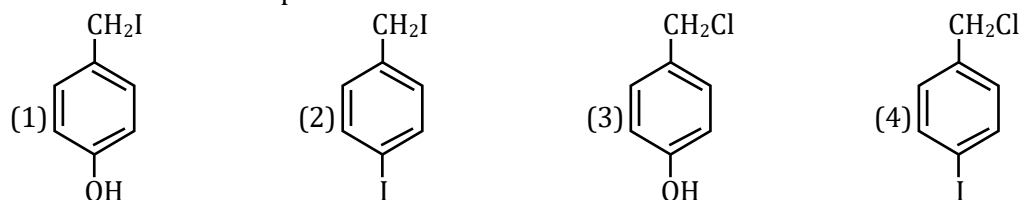


CRM105

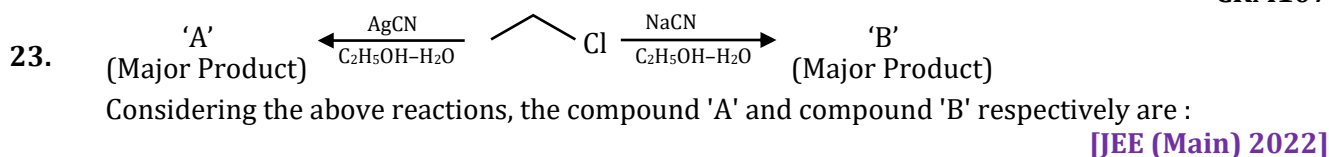
CRM106



In the above reaction product B is:



CRM107



CRM108

24. The correct pair(s) of the ambident nucleophiles is (are) :
 (A) AgCN/KCN (B) RCOOAg/RCOOK
 (C) AgNO₂/KNO₂ (D) AgI/KI
 (1) (B) and (C) only (2) (A) only (3) (A) and (C) only (4) (B) only

[JEE (Main) 2021]

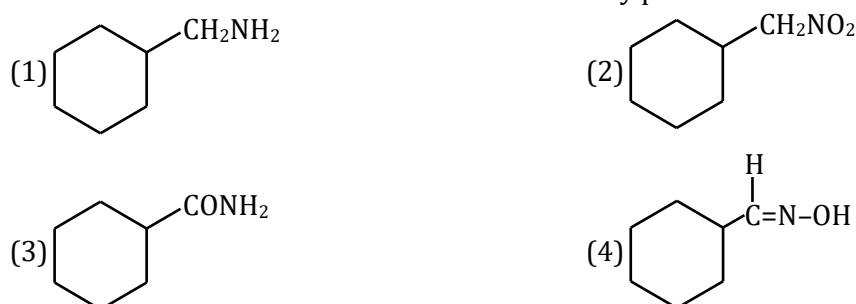
25. Given below are two statements :
Statement I : C₂H₅OH and AgCN both can generate nucleophile.
Statement II : KCN and AgCN both will generate nitrile nucleophile with all reaction conditions.
 Choose the most appropriate option :
 (1) Statement I is true but statement II is false
 (2) Both statement I and statement II are true
 (3) Statement I is false but statement II is true
 (4) Both statement I and statement II are false

CRM109

[JEE (Main) 2021]

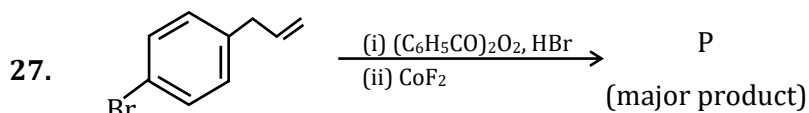


Consider the above chemical reaction and identify product "A"



CRM110

CRM111



Major product P of above reaction, is :

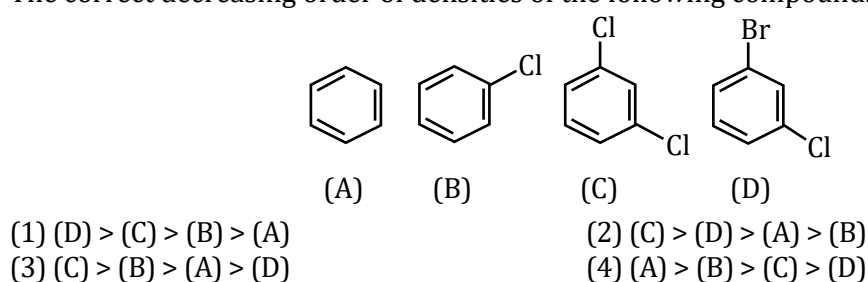
[JEE (Main) 2021]



28. The correct decreasing order of densities of the following compounds is :

CRM112

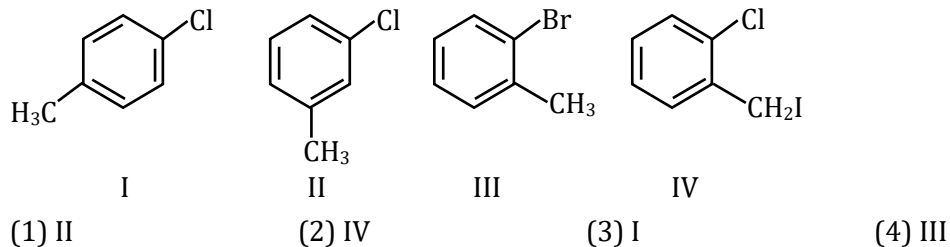
[JEE (Main) 2021]



29. Among the following compounds I-IV, which one forms a yellow precipitate on reacting sequentially with (i) NaOH (ii) dil. HNO₃ (iii) AgNO₃?

CRM113

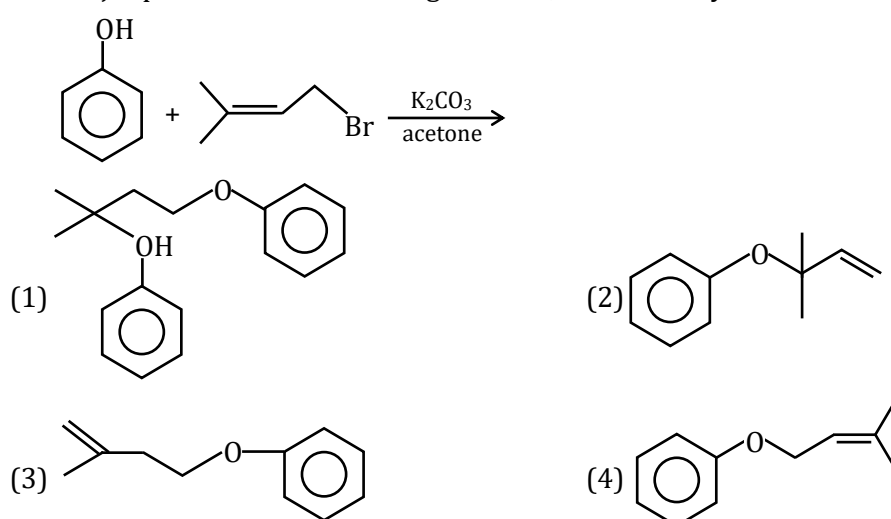
[JEE (Main) 2021]



30. The major product of the following reaction, if it occurs by S_N2 mechanism is :

CRM114

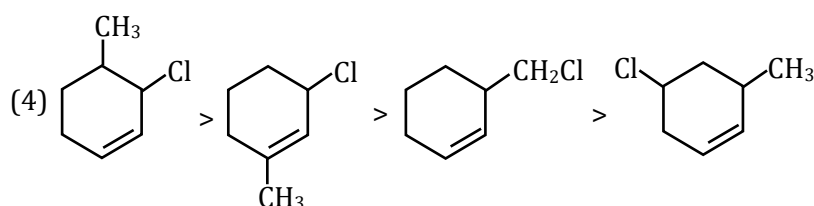
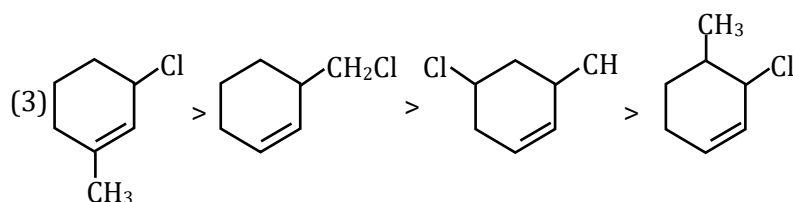
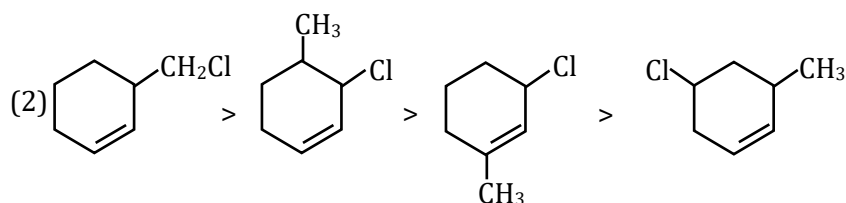
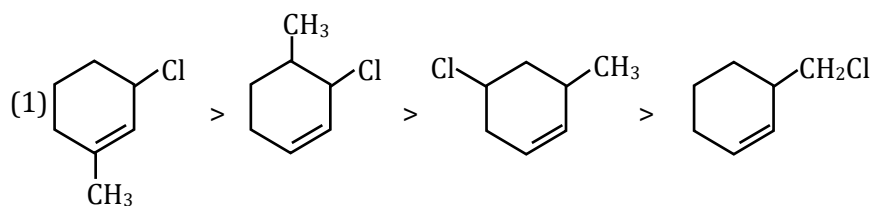
[JEE (Main) 2021]



CRM115

31. The **correct** order of reactivity of the given chlorides with acetate in acetic acid is :

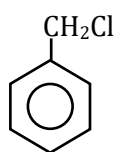
[JEE (Main) 2021]



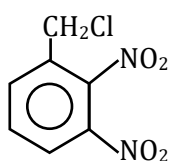
CRM116

32. The decreasing order of reactivity of the following compounds towards nucleophilic substitution (S_N2) is :

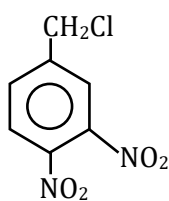
[JEE (Main) 2020]



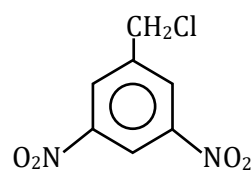
(I)



(II)



(III)



(IV)

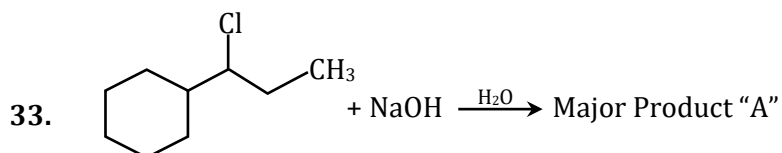
(1) (IV) > (II) > (III) > (I)

(2) (II) > (III) > (IV) > (I)

(3) (II) > (III) > (I) > (IV)

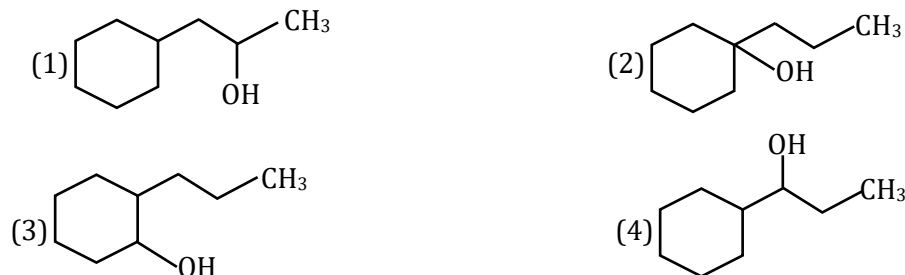
(4) (III) > (II) > (IV) > (I)

CRM117



Consider the above chemical reaction. Product "A" is:

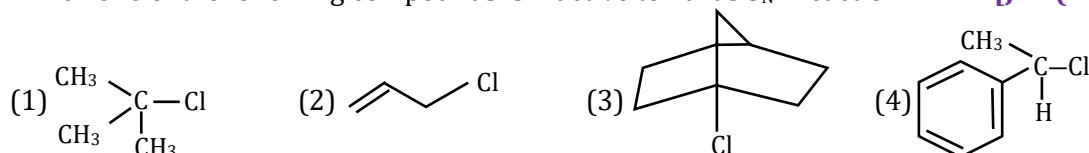
[JEE (Main) 2024]



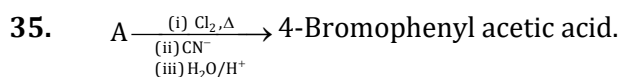
CRM118

34. Which one of the following compounds is inactive towards $\text{S}_{\text{N}}1$ reaction?

[JEE (Main) 2022]

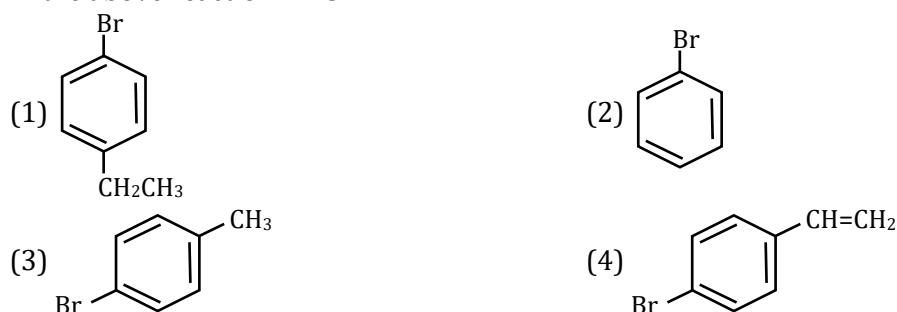


CRM119



[JEE (Main) 2022]

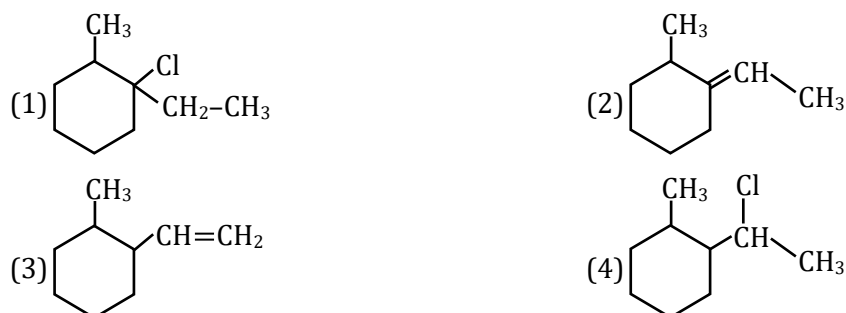
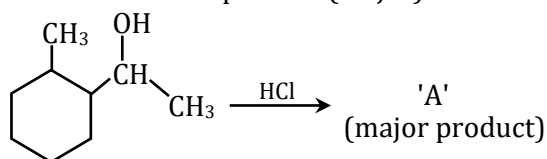
In the above reaction 'A' is



CRM120

36. What is the final product (major) 'A' in the given reaction ?

[JEE (Main) 2021]



CRM121

37. In the given reaction 3-Bromo-2, 2-dimethyl butane $\xrightarrow{\text{C}_2\text{H}_5\text{OH}}$ 'A' Product A is:
(Major Product)

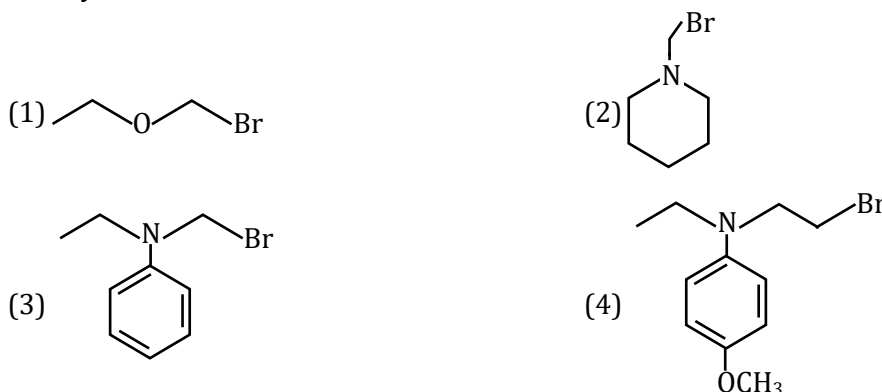
[JEE (Main) 2021]

- (1) 2-Ethoxy-3, 3-dimethyl butane
(2) 1-Ethoxy-3, 3-dimethyl butane
(3) 2-Ethoxy-2, 3-dimethyl butane
(4) 2-Hydroxy-3, 3-dimethyl butane

CRM122

38. Which of the following compounds will form the precipitate with aq. AgNO_3 solution most readily ?

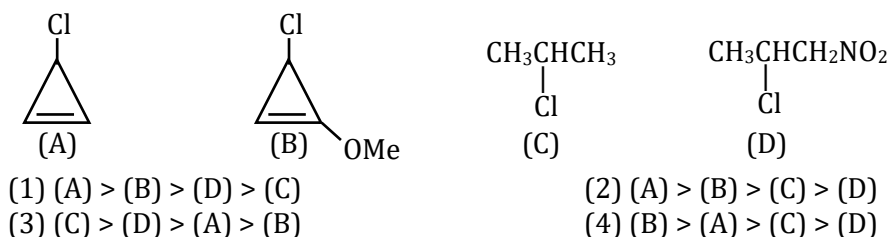
[JEE (Main) 2020]



CRM123

39. The decreasing order of reactivity of the following organic molecules towards AgNO_3 solution is :

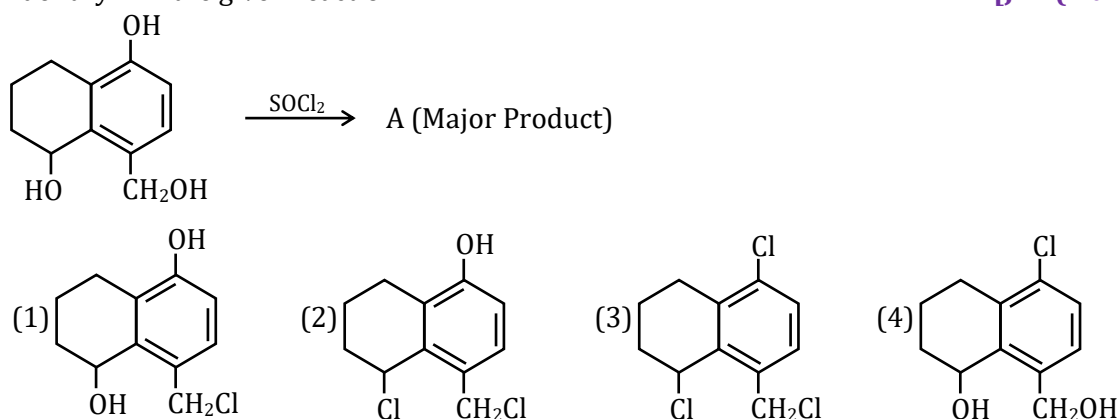
[JEE (Main) 2020]



CRM124

40. Identify A in the given reaction.

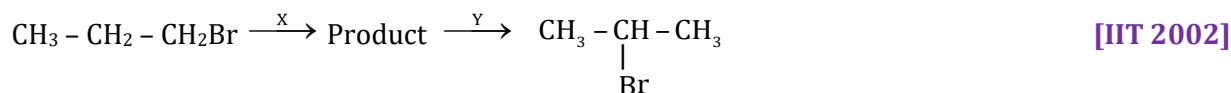
[JEE (Main) 2021]



CRM125

EXERCISE - JEE (Advanced) PYQ

1. Identify the set of reagents / reaction conditions 'X' and 'Y' in the following set of transformation:



- (A) X = dilute aqueous NaOH, 20°C; Y = HBr / acetic acid, 20°C
(B) X = concentrated alcoholic NaOH, 80°C; Y = HBr / acetic acid 20°C
(C) X = dilute aqueous NaOH, 20°C; Y = Br₂ / CHCl₃, 0°C
(D) X = concentrated alcoholic NaOH, 80°C; Y = Br₂/CHCl₃, 0°C

CRM126

2. An S_N2 reaction at an asymmetric carbon of a compound always gives. [IIT 2001]

- (A) an enantiomer of the substance
(B) a product with opposite optical rotation
(C) a mixture of diastereomers
(D) a single stereoisomer

CRM127

3. The compound that will react most readily with NaOH to form methanol is [IIT 2001]

- (A) (CH₃)₄N⁺I[−] (B) CH₃OCH₃ (C) (CH₃)₃S⁺I[−] (D) (CH₃)₃CCl

CRM128

4. The order of reactivity of the following alkyl halides for a S_N2 reaction is: [IIT 2000]

- (A) R-F > R-Cl > R-Br > R-I (B) R-F > R-Br > R-Cl > R-I
(C) R-Cl > R-Br > R-F > R-I (D) R-I > R-Br > R-Cl > R-F

CRM129

5. Which of the following has the highest nucleophilicity? [IIT 2000]

- (A) F[−] (B) OH[−] (C) CH₃[−] (D) NH₂[−]

CRM130

6. The products of reaction of alcoholic AgNO₂ with ethyl bromide are [IIT 1991]

- (A) Ethane (B) Ethyl nitrite (C) Nitroethane (D) Ethyl alcohol

CRM131

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	B	A	A	B	A	B	C	B	D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	C	C	D	A	C	D	B	D	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	D	B	D	C	A	A	C	D	D
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	D	D	B	C	A	A	B	D	B	B
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	D	B	C	C	C	B	B	A	B	C
Que.	51	52	53	54	55	56				
Ans.	A	C	D	D	C	B				

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,C	B,C,D	B,D	A,B	A,B,D	B,D	A,B	A,B,C	C,D	C,D
Que.	11									
Ans.	A,B,C,D									

EXERCISE (S)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	5	3	2	4	4	6	4	12	3
Que.	11	12	13	14	15	16	17	18		
Ans.	5	6	5	3.00	4.00	4.00	3.00	8.00		

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	3	3	3	2	2	4	4	2	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	3	3	1	4	4	2	2	2	1
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	2	1	3	3	1	3	4	1	2	4
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	1	2	2	3	3	1	3	2	4	2

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6				
Ans.	B	D	A	D	C	C				